



OSI & TCP/IP Models

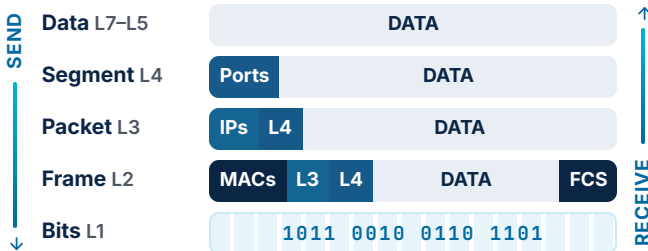
Use this when: you troubleshoot layer by layer, or need to place a protocol, device or attack at its layer.

7 OSI layers ↔ 4 TCP/IP layers

Sender works 7 → 1 · receiver works 1 → 7

TCP/IP	#	OSI LAYER	WHAT IT DOES	PDU	KEY PROTOCOLS	DEVICE · ADDRESS
Application L5-L7	7	Application	Network services for apps	Data	HTTP(S), DNS, DHCP, SSH	Host, server, NGFW
	6	Presentation	Format, encrypt, compress	Data	TLS*, UTF-8, JPEG	Endpoint
	5	Session	Open & close sessions	Data	RPC, NetBIOS	Endpoint
Transport	4	Transport	End-to-end delivery, ports	Segment	TCP, UDP	Firewall · port
Internet	3	Network	Routing between networks	Packet	IPv4, IPv6, ICMP, OSPF	Router · IP address
Link L1-L2	2	Data Link	Frames, MAC, error check	Frame	Ethernet, Wi-Fi, ARP*	Switch, AP · MAC
	1	Physical	Signals: voltage, light, radio	Bits	1000BASE-T, fiber, PoE	Cable, hub, NIC

Encapsulation



TCP vs UDP

Layer 4

	TCP	UDP
SETUP	SYN → SYN-ACK → ACK	None
DELIVERY	ACKs, retransmit, windowing	Best effort
HEADER	20 bytes min.	8 bytes
USED BY	HTTPS 443, SSH 22	DNS 53*, DHCP 67/68

Troubleshoot bottom-up

Cisco IOS · Windows

- 1 `show interfaces status` errors, duplex
- 1 `shutdown` → `no shutdown` **CAUTION** bounce port
- 2 `show mac address-table` port & VLAN?
- 3 `show ip interface brief` up/up, IP?
- 3 `ping` · `tracert` · `tracert` path breaks?
- 4 `Test-NetConnection 192.0.2.10 -Port 443` port open?
- 7 `nslookup example.com` resolves?

CAUTION `shutdown` drops every session on that port. Never bounce the link you are managing from.

Attacks & defenses by layer

- 7 Phishing, SQL injection Email filter, WAF
- 4 SYN flood, port scans SYN cookies, firewall
- 3 IP spoofing ACLs, uRPF
- 2 ARP spoofing Dynamic ARP Inspection
- 2 MAC flooding, rogue DHCP Port security, DHCP snooping
- 1 Rogue device on a jack 802.1X, shut unused ports

AUTHORIZED USE ONLY Test attacks only in your own lab or with written permission.

REMEMBER

- L1 → L7: Please Do Not Throw Sausage Pizza Away
- L7 → L1: All People Seem To Need Data Processing
- PDUs: Data → Segment → Packet → Frame → Bits

- Switch = L2 (MAC) · Router = L3 (IP)
- UDP's L4 PDU is a **datagram**; DNS also uses TCP 53*
- * ARP sits between L2/L3; TLS is often placed at L6